AJAN MUTHURAJ

ROBOTICS & AUTONOMOUS SYSTEMS ENGINEER

ajanm2003@gmail.com
github.com/AjanM27 | linkedin.com/in/ajanmuthuraj
ajanm27.github.io/ajan-portfolio

01 PROFILE

IIT Madras Mechanical Engineering graduate building end-to-end robotic systems across safe reinforcement learning, aerial autonomy, multi-robot mapping, motion planning, perception and manipulation. Strongest at turning research ideas into observable, reproducible systems that connect simulation, algorithms and deployment.

02 EDUCATION

Indian Institute of Technology Madras
2022 - 2026

B.Tech, Mechanical Engineering | CGPA: 8.52/10

Thesis: Reach-Avoid Reinforcement Learning with Time-Varying Safe and Target Zones.

03 EXPERIENCE & RESEARCH

Robotics Research Intern - IIT Roorkee
Jun - Jul 2025

Humanoid navigation, perception and manipulation

- Integrated Gazebo/URDF simulation, SLAM localization and Nav2 autonomous navigation for a hospital-like environment.
- Built YOLOv8 and depth-based 3D localization, then connected MoveIt 2 and OctoMap for collision-aware manipulation; reported approximately 96.6% grasp success across 30 trials.

Systems Engineer - Dai-ichi Life Techno Cross
Upcoming 2026

Japan

- Incoming systems-engineering role following graduation.

04 SELECTED PROJECTS

Reach-Avoid RL in Time-Varying Environments
Nov 2025 - May 2026

B.Tech thesis | PyTorch, DDQN, Gym

- Extended the discounted reach-avoid formulation with time-augmented state, corrected baseline implementation issues and trained a DDQN for moving obstacles and targets; reported up to 94.4% safe-state prediction precision.

Autonomous UAV Exploration in Isaac Sim
2026

Isaac Sim, ROS 2, ArduPilot, RTAB-Map

- Built and documented a reproducible stack from simulated sensors and flight control through SLAM, collision-aware 3D frontier exploration and multi-drone launch workflows.

Multi-Robot Exploration & Temporal Map Fusion
2025 - 2026

Collaborative research | SLAM, map fusion, control

- Worked on shared occupancy-grid memory, deadlock recovery and coordinated frontier exploration under noisy sensing; wider work accepted as an ICRA 2026 XPLORE Workshop poster.

Dynamic Path Planning
Apr 2025

Adaptive RRT*, LPA*, ROS 2, Gazebo

- Implemented and benchmarked planners in moving-obstacle scenarios; reported approximately 12% shorter paths on average in the evaluated cases.

05 TECHNICAL SKILLS

ROBOTICS	ROS 2, Nav2, MoveIt 2, Gazebo, Isaac Sim, RViz, ArduPilot, RTAB-Map, SLAM Toolbox
AI / PLANNING	PyTorch, DDQN, TD3, YOLOv8, OpenCV, RRT*, LPA*, frontier exploration
ENGINEERING	Python, C++, MATLAB, Linux, Git, Docker, Arduino, ESP32, system integration

06 RECOGNITION & CREDENTIALS

ICRA 2026 XPLORE Workshop: Poster acceptance for collaborative work on multi-robot frontier-based exploration under uncertainty with SLAM and temporal map fusion.
Certifications: Aerial Robotics - University of Pennsylvania (2025); Robotics Software Engineer Nanodegree - Udacity (2024); e-Yantra Robotics Competition - IIT Bombay (2024).